

Geometry-aware view-dependent 3D ultrasound implicit representation reconstruction

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ABSTRACT

Reconstruction of three-dimensional (3D) scenes from a set of two-dimensional ultrasound images has garnered significant attention in medical research nowadays. Recent studies have demonstrated that neural implicit representation provides a novel and high-fidelity solution. However, most of the existing implicit methods ignore the strong physical correlation between ultrasound reflections and the geometry of human tissues, limiting their ability to distinguish between reflections from large tissue interfaces and scattering from microstructures. Moreover, the existing methods do not take into account the scale inconsistency between features of tissues and organs, resulting in loss of detailed information. In this work, we propose a geometry-aware view-dependent implicit 3D ultrasound reconstruction method. First, we design a novel physics-based volume rendering model that can reconstruct anisotropic and geometry-related reflections more accurately. Then, we employ a multi-resolution hash encoding strategy to capture features at various scales for both tissues and organs, which not only helps differentiate between broader reflections and fine-grained scattering but also significantly accelerates our reconstruction process. Experiments demonstrate that the proposed method achieves high visual quality and rapid 3D ultrasound reconstruction. Moreover, additional experiments demonstrate the superior performance of the proposed method in distinguishing ultrasound reflections from scattering while preserving the anisotropy of reflections, thereby assisting medical personnel in effectively identifying tissue structures.

1. Introduction

The majority of two-dimensional (2D) ultrasound devices are limited to generating single cross-sectional images of anatomical structures, which presents sonographers with the complex challenge of mentally assembling a 3D scene from 2D ultrasound scan data. In addition, due to the anisotropic characteristics of ultrasound imaging data, the aforementioned difficulty is further exacerbated. Therefore, 3D reconstruction techniques based on 2D ultrasound imaging data have attracted more and more attention in medical field. Nevertheless, the current methodologies for explicit three-dimensional ultrasonographic reconstruction [1–3], which use low-resolution tessellated grids, are constrained by the inherent limitations of finite resolution and suffer from the loss of continuity [4].

In recent years, implicit neural representation methods [5–11], e.g., neural radiance field (NeRF) [6], have demonstrated impressive

performance in 3D scene reconstruction and novel view synthesis from a set of multi-view images. These methods offer a promising alternative for 3D ultrasound reconstruction. For example, Li et al. [9] and Song et al. [10] respectively apply the NeRF algorithm for 3D ultrasound image reconstruction of spine data and carotid vessels. However, these methods do not account for the physical properties of ultrasound, specifically the intrinsic differences in appearance and geometry that arise with changes in viewing angles, leading to inaccuracies in reconstruction. Ultra-NeRF [11] tries to solve the above problem by introducing a physics-based rendering equation into the implicit neural representation to simulate the mechanics of ultrasound imaging. Specifically, they propose that 2D ultrasound imaging data can be decomposed into distinct reflection and scattering components. This decomposition can be achieved by analyzing tissue properties derived from overlapping ultrasound sweeps, allowing the model to learn the unique contributions of each component. Ultra-NeRF [11] makes a great progress in 3D implicit reconstruction of

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ultrasound by accounting for the physical priors. However, it still suffers from the following issues. First, Ultra-NeRF ignores the strong physical correlation between ultrasound reflections and the geometry of human tissues, limiting its ability to distinguish between reflections from large tissue interfaces and scattering from microstructures. Second, it does not take into account the scale inconsistency between features of tissues and organs, resulting in loss of detailed information.

To address the aforementioned challenges, this paper proposes a geometry-aware view-dependent implicit 3D ultrasound reconstruction method. First, we design a novel volume rendering model specifically adapted to ultrasound physics. To address the anisotropic nature of ultrasound reflection, we leverage implicit neural networks with high representation capacity to model tissue properties for volume rendering, particularly focusing on acoustic impedance. Additionally, the proposed rendering model is geometry-aware, enabling direct calculation of surface normals for tissue cross-sections by leveraging the neural gradient of acoustic impedance. Then, to further distinguish reflections from scattering, we implement a multi-scale hash encoding strategy to capture tissue features across various scales. The basic motivation is that coarse-scale hash encoding is particularly beneficial for optimizing reflections, as these reflections often span large tissue cross-sections. Additionally, coarse-scale encoding effectively mitigates scattering noise interference within tissues. Meanwhile, multi-scale hash encoding enables the model to capture fine-grained scattering details. Furthermore, hash encoding strategy can significantly accelerate our training and reconstruction process. Experimental results indicate that the proposed implicit representation model can reconstruct a 3D ultrasound scene with higher geometric accuracy and more precise physical property estimation, significantly improving its capacity to differentiate between reflections and scattering.

In summary, the main contributions of the proposed method include:

- A novel geometry-aware view-dependent 3D ultrasound neural implicit representation method is proposed, enabling more accurate and high-fidelity ultrasound reconstruction.
- A new volume rendering equation specifically adapted to ultrasound physics is proposed, enhancing the distinction between reflections and scattering through surface normal estimation.
- A multi-scale hash encoding strategy is employed to capture tissue features at various scales, further enhancing the ability to distinguish reflections from scattering while also accelerating both the training and reconstruction processes.
- Qualitative and quantitative experiments demonstrate the significant advantages of the proposed approach.

2. Related work

2.1. Neural implicit representation

While many forms of 3D shape representation are used in deep learning based geometry processing, such as point clouds [12,13], meshes [14,15], and voxel grids [16–18], neural implicit functions [6, 19–21] have attracted growing attention from researchers for their ability to represent complex topologies and achieve theoretically infinite resolution. In the original NeRF [6], a scene is represented by a fully-connected neural network. This network takes a spatial position and view direction as inputs and outputs the volume density and radiance at that location. The final color for each pixel is computed using the following formula:

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\sigma(t)c(t)dt \quad (1)$$

where $\mathbf{r}$ defines a ray, $T(t)$ represents transmittance factor, $\sigma(t)$ and $c(t)$ are the volume density, and the radiance at position t respectively. The integral accumulates the ray-traced radiance field from the near plane t_n to the far plane t_f , resulting in the final pixel color. This color is obtained by weighting the color of each sampling point along the ray.

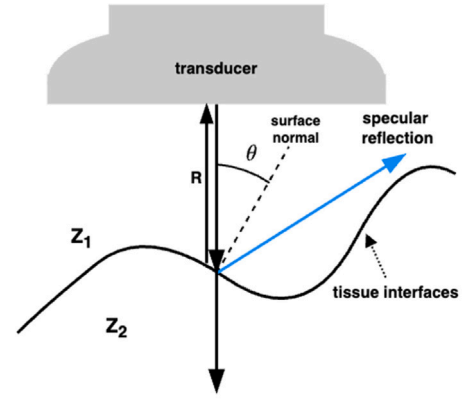


Fig. 1. Principle of ultrasound transmission and reflection at multiple tissue interfaces.

More details can be found in paper [6]. It is worth noting that in order to obtain high-frequency detail, the input spatial position is first encoded by Fourier transformation, and then fed into the fully-connected neural network.

Training a NeRF is computationally intensive, as it requires extensive sampling throughout the entire scene space. Therefore, to accelerate the training process of NeRF, some improvements have been proposed in [22–24]. Instant-NGP [24] introduces a multi-resolution hash encoding within voxel grids, demonstrating significant improvements in training speed. In detail, the voxel grids at each resolution are mapped to a hash table with fixed-size learnable features. The final hash encoding is obtained by concatenating these learnable features at all scales:

$$h_{\Omega}(\mathbf{x}) = (h_{\Omega}^1(\mathbf{x}), h_{\Omega}^2(\mathbf{x}), \dots, h_{\Omega}^L(\mathbf{x})) \quad (2)$$

where $h_{\Omega}^l(\mathbf{x})$ with $l \in \{1, \dots, L\}$ denotes the hash encoding of the coordinate $\mathbf{x}$ at spatial scale level l . Instant-NGP replaces the original NeRF's positional encoding with the final hash encoding output, significantly accelerating training speed while preserving the model's high-frequency information fitting capability.

2.2. 3D ultrasound reconstruction

In this section, we will introduce the basic theory of ultrasound physics, and more details can be found in [25,26]. In ultrasound imaging, high-frequency sound waves are emitted by the ultrasound transducer and propagate through human tissues. When these waves encounter tissues with different acoustic properties, they are reflected and scattered. These echo waves are then detected by the transducer and used to create detailed images of the internal structures. In general, the physical mechanism behind ultrasound image synthesis can be viewed as a combination of two key components: reflection and scattering,

$$E(\mathbf{r}, t) = R(\mathbf{r}, t) + B(\mathbf{r}, t) \quad (3)$$

where $\mathbf{r}$ denotes the direction of the scanning line, while $R(\mathbf{r}, t)$ is the reflected intensity from the tissue boundaries and $B(\mathbf{r}, t)$ is the scattering intensity from the scattering points.

Reflection occurs when an incident ultrasound pulse encounters a large interface of two body tissues with different acoustic impedance, such as the diaphragm and the kidney surface. In general, as shown in Fig. 1, the echo intensity of reflection relates to the angle θ between the incidence and the surface normal, as well as the difference in the acoustic impedance z_1, z_2 of two adjacent tissues. In accordance with Lambert's cosine law, the reflected energy can be simulated as [27]:

$$R(\mathbf{r}, t) = \left| I(\mathbf{r}, t) * \cos \theta^n * \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2 \right| * H(\mathbf{r}, t) \otimes G(x, y, z) \quad (4)$$

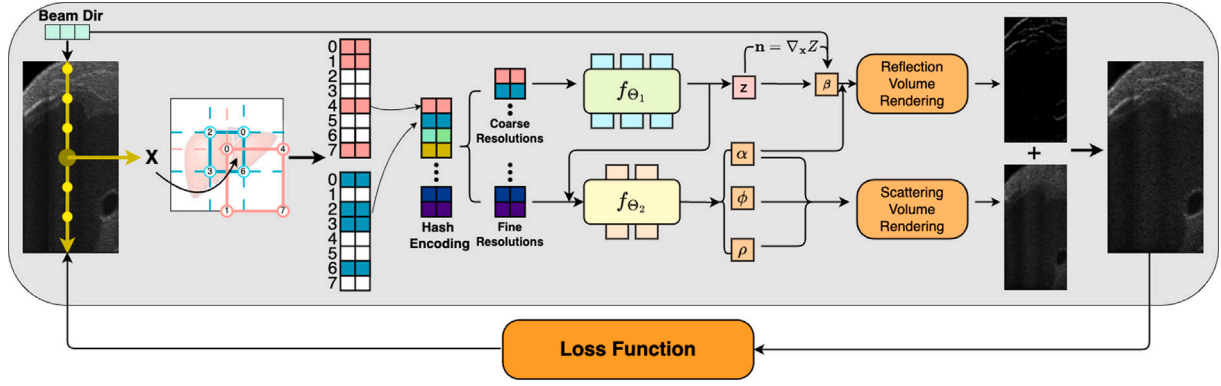


Fig. 2. Geometry-aware view-dependent 3D ultrasound reconstruction framework. Given a spatial point x , the extracted coarse-scale hash encoding features are used to learn the reflection and the fine-scale hash encoding features are utilized to learn the fine-grained scattering.

where $I(\mathbf{r}, t)$ is the remaining ultrasound wave amplitude at the distance t , $H(\mathbf{r}, t)$ is the PSF of the imaging system, $G(x, y, z)$ is an indicator function that returns 1 for points on the surface boundaries and 0 otherwise and the exponent n is a simple modification to describe the heterogeneity on the surface.

Scattering happens when a sound wave interacts with a structure that has a different acoustic impedance from the surrounding tissue and is smaller than the wavelength of the incident sound wave, such as red blood cells or glomeruli. In general, scattering intensity is much weaker than reflection intensity and is largely independent of the incident sound wave's direction. According to [27], the scattering intensity can be defined as:

$$B(\mathbf{r}, t) = I(\mathbf{r}, t) * H(\mathbf{r}, t) \otimes T(x, y, z) \quad (5)$$

where $T(x, y, z)$ represents the scatterers generated by a generative model.

3D ultrasound reconstruction aims to create a three-dimensional representation of the scene by combining multiple 2D images captured from scans taken along different angles or directions. Once the 3D ultrasound representation is reconstructed, doctors can examine the scene from various angles without the need for additional complex scanning, thereby easing the burden on both doctors and patients. Conventional ultrasound reconstruction methods [1–3,28–30] are inherently limited by the discrete voxel grid, resulting in a lack of continuity. For example, Sherebrin et al. [2] uses the nearest pixel from the source 2D image [4] to replace the value in each voxel within the 3D volume. Benefiting from the high-fidelity 3D representation capabilities of NeRF, researchers are exploring ways to extend its applications into the medical field. MedNeRF [31] focuses on learning a continuous representation of CT scans from a few single-view X-rays by disentangling the shape and volumetric depth of surface and internal anatomical structures from 2D images. Li et al. [32] and Zaidkilani et al. [33] have also made pioneering attempts in addressing issues in the field of ultrasound through deep learning methodologies. Li et al. [9] and Song et al. [10] apply the NeRF algorithm for 3D ultrasound image reconstruction of spine data and carotid vessel, respectively. However, these methods fail to account for the physical properties of ultrasound, particularly the inherent changes in appearance and geometry that occur with variations in viewing angle, which can result in inaccurate reconstructions.

The most recent and related work, Ultra-NeRF [11], attempts to address this issue by introducing an implicit neural representation with physics-based rendering for 3D ultrasound reconstruction. Ultra-NeRF propose to estimate the reflectance of each sample point along the scan line,

$$R(\mathbf{r}, t) = |I(\mathbf{r}, t) \cdot \beta(\mathbf{r}, t)| \cdot PSF(\mathbf{r}) \otimes G(\mathbf{r}', t') \quad (6)$$

where $\beta(\mathbf{r}, t)$ represents the reflectance, $PSF(\mathbf{r})$ is a predefined 2D point-spread function, and $G(\mathbf{r}, t)$ admits 1 for points at the boundary and 0 otherwise. Similar to the original NeRF, Ultra-NeRF employs a multi-layer perception (MLP) network to map a position x to tissue parameters, such as reflectance β .

Ultra-NeRF incorporates the physical principles of ultrasound into 3D implicit representation, which enhances reconstruction performance to some extent. However, it is evident that, compared to (4), Ultra-NeRF disregards the anisotropic reflection mechanisms at tissue interfaces between distinct acoustic impedance regions, instead directly optimizing a position-dependent reflectance β that ignores acoustic wave directionality. As a result, Ultra-NeRF struggles to accurately reconstruct the geometry of tissue interfaces, leading to confusion between reflection and scattering, and ultimately degrading its performance in anisotropic synthesis across different viewpoints. Some examples are shown in Fig. 3 and Fig. 6.

3. Main method

In this section, we will introduce the proposed geometry-aware ultrasound reconstruction method in detail. The pipeline of the proposed method is shown in Fig. 2. Unlike Ultra-NeRF, which directly predicts the reflectance, the proposed method better adheres to the physical principles of ultrasound. We propose a novel geometry-aware ultrasound rendering framework through predicting the acoustic impedance of each 3D point within the space, yielding more accurate 3D reconstructions with explicit geometry constraints. Additionally, we observe that reflections are more influenced by coarse-scale features, whereas scattering is predominantly determined by fine-scale details. Therefore, we propose a multi-scale hash encoding strategy that not only effectively distinguishes reflections from scattering but also significantly accelerates the training process. A detailed explanation of the main method will be presented in the following sections.

3.1. Separable multi-scale hash encoding

Due to the physical mechanism of ultrasound imaging, reflections occur when an incident ultrasound pulse encounters a large interface of two body tissues with different acoustic impedance. Therefore, the reflection intensity is primarily determined by the contrast in acoustic impedance between the tissues, which corresponds to coarser-scale features. In contrast, scattering occurs when a sound wave interacts with a structure that has a different acoustic impedance than the surrounding tissue and is smaller than the wavelength of the incident wave, making it more closely associated with fine-scale features.

Based on the above analysis, we propose a separable multi-scale hash encoding strategy in this section. Given a 3D position $\mathbf{x} = (x, y, z)$ in Cartesian space, we construct a separable multi-scale feature embedding

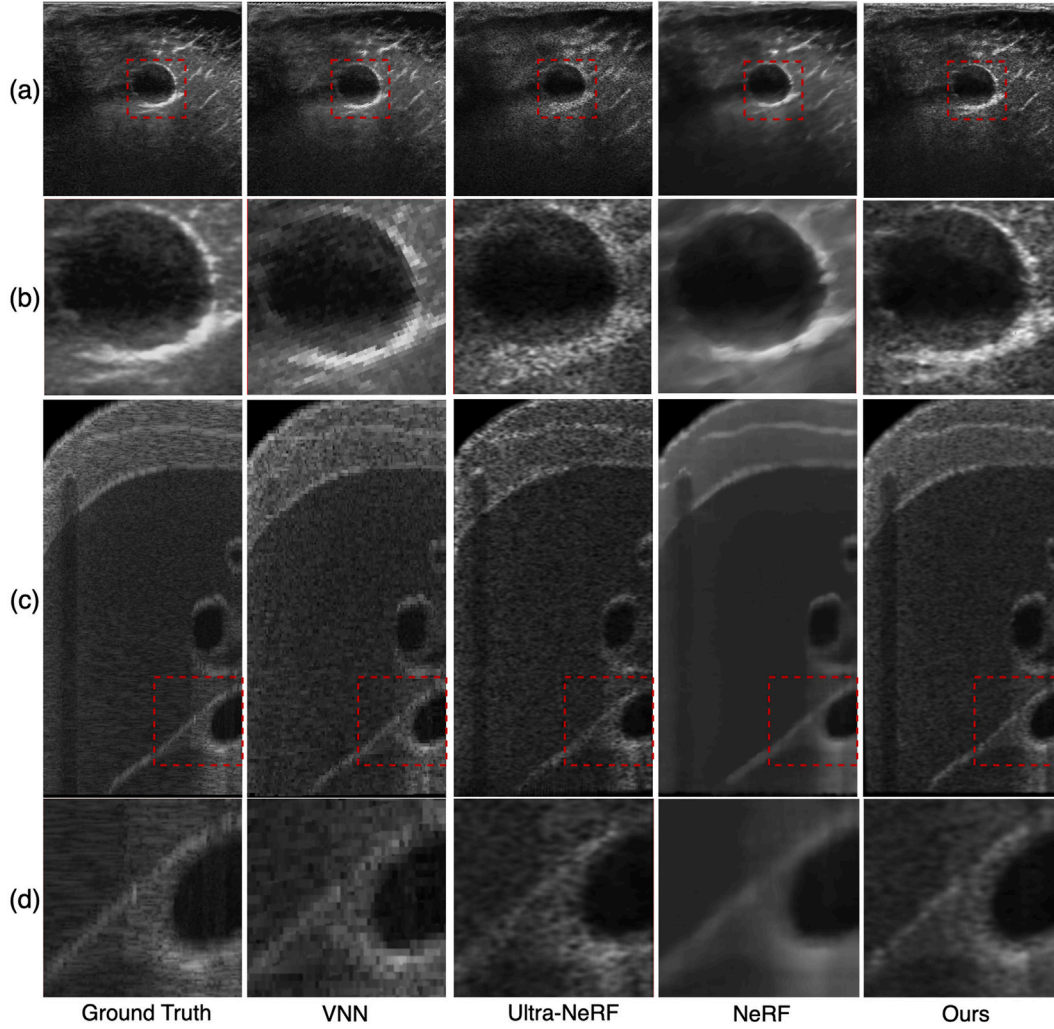


Fig. 3. Qualitative comparisons of different methods for test views on the pork dataset and the liver dataset.

using hash encoding, which yields both coarse-scale and fine-scale features at position $\mathbf{x}$. On the one hand, hash encoding provides a compact and memory-efficient representation, which accelerates network convergence. On the other hand, by leveraging the spatial-scale encoding capability of multi-scale hash encoding, our method decouples coarse and fine spatial information, thereby enhancing the model's ability to distinguish between reflections and scattering.

For the 3D position $\mathbf{x}$ in a scan line, we map it to its each scale hash encodings $h_{\Omega}^i(\mathbf{x}) \in \mathbb{R}^d$ ($i = 1, \dots, L$) using learnable hash table entries Ω , where d denotes the feature dimension of $h_{\Omega}^i(\mathbf{x})$ and L is the number of spatial scale. The hash encodings at all L scales are then concatenated to be the multi-resolution hash encoding $h_{\Omega}(\mathbf{x}) = \{h_{\Omega}^i(\mathbf{x})\}_{i=1}^L$. As $h_{\Omega}(\mathbf{x})$ provides a multi-scale encoding of spatial position, it efficiently captures acoustic impedance and other tissue parameters. In this paper, we set $L = 16$ to achieve both high fidelity and efficient reconstruction:

$$h_{\Omega}(\mathbf{x}) = (h_{\Omega}^1(\mathbf{x}), \dots, h_{\Omega}^{16}(\mathbf{x})). \quad (7)$$

Specifically, we take the first L_1 scales as the coarse-scale features:

$$h_{\Omega_1}(\mathbf{x}) = \{h_{\Omega}^i(\mathbf{x})\}_{i=1}^{L_1}, \quad (8)$$

and the remaining as the fine-scale features:

$$h_{\Omega_2}(\mathbf{x}) = \{h_{\Omega}^i(\mathbf{x})\}_{i=L_1+1}^{16}. \quad (9)$$

In our experiments, we set $L_1 = 8$, which yields effective decoupling of spatial features and robust reconstruction accuracy. Next, the coarse-scale hash encoding $h_{\Omega_1}(\mathbf{x})$ and fine-scale hash encoding $h_{\Omega_2}(\mathbf{x})$ are fed into Neural Network f_{Θ_1} and f_{Θ_2} , respectively, to predict the reflection and scattering intensities at that position $\mathbf{x}$.

3.2. Geometry-aware ultrasound rendering

To ensure better adherence to the physical principles of ultrasound, we propose a novel geometry-aware ultrasound rendering framework. Rather than directly regressing the position-dependent reflectance via an MLP network, as in Ultra-NeRF, we instead propose to learn the acoustic impedance z of different tissue position, allowing for the direct calculation of the surface normal $\mathbf{n}$ at tissue interfaces. From there, the reflectance β can be computed using Lambert's cosine law.

As acoustic impedance characterizes the global properties of tissues, we propose to predict the acoustic impedance feature z of each position via a neural implicit function f_{Θ_1} with the coarser-scale feature serving as input,

$$(z, \mathbf{g}) = f_{\Theta_1}(h_{\Omega_1}(\mathbf{x})) \quad (10)$$

where, f_{Θ_1} is an MLP network, $\mathbf{g}$ represents a learned geometry feature derived from the coarse-scale hash encoding features $h_{\Omega_1}(\mathbf{x})$.

Since acoustic impedance governs the reflection characteristics of different tissues, its gradient with respect to the coordinate $\mathbf{x}$ can be

approximated as the normal vector to the interface between the two tissues. Therefore, in this paper we propose to calculate the normal vector via $\mathbf{n} = \frac{\nabla_{\mathbf{x}} z}{|\nabla_{\mathbf{x}} z| + \epsilon}$. This normalization strategy effectively decouples the geometric direction from the gradient magnitude, significantly alleviating instability issues caused by drastic magnitude fluctuations near discontinuities. Additionally, to prevent numerical singularities in regions where the gradient magnitude approaches zero, we implement a smooth normalization by adding a small constant ϵ (10^{-6}) to the denominator. Furthermore, the reflection parameter $\beta(\cdot)$ as defined in (6) and (4) can be derived directly from Lambert's cosine law,

$$\beta = \left[\left(\frac{\nabla_{\mathbf{x}} z}{|\nabla_{\mathbf{x}} z| + \epsilon} \right)^T \mathbf{d} \right]^2 * \left(\frac{z_1 - z_2}{z_1 + z_2} \right)^2 \quad (11)$$

where $\mathbf{d}$ is the scan-line direction of the ultrasound wave, and z_1 and z_2 correspond to the acoustic impedance of the adjacent tissues. Crucially, this formulation establishes a synergy between explicit physics-based modeling and the implicitly learned acoustic impedance field. By enforcing Lambert's cosine law using the surface normal derived from the impedance gradient ($\nabla_{\mathbf{x}} z$), the model explicitly governs the mechanism of view-dependence, ensuring that reflection intensity strictly follows the geometry determined by physical laws rather than a black-box mapping.

Therefore, after taking attenuation into account [34], the reflection $R(\mathbf{r}, t)$ can be defined as:

$$R(\mathbf{r}, t) = I(\mathbf{r}, t) * PSF(\mathbf{r}, t) \otimes \beta(\theta, z_1, z_2) \quad (12)$$

where, β is a reflectance function as defined in (11) that depends on θ, z_1, z_2 and $PSF(\mathbf{r}, t)$ represents a predefined 2D point-spread function. The attenuation of $I(\mathbf{r}, t)$ can be attributed to the absorption of ultrasonic waves by human tissues, reflection at tissue boundaries, and attenuation compensation via a single attenuation coefficient α , so it is modeled as $I(\mathbf{r}, t) = I_0 e^{-\int_{t_0}^t \alpha dt}$, where I_0 is the initial intensity of ultrasound and α is the attenuation coefficient of the medium. Specifically, when performing the integration along the ray (12), we map the time-of-flight to spatial depth using a constant speed of sound, effectively treating the ultrasound propagation path as a straight line and ignoring the geometric distortions caused by refraction. Unlike the formula in Ultra-NeRF (6), we improve the computation of the reflectance based on ultrasound-physical principles; furthermore, given the strong correlation between tissue boundaries and reflectance, we remove the boundary indicator function and instead directly calibrate boundaries by the magnitude of the reflection coefficient, which further prevents overfitting of ultrasound reflections.

Distinct from directly predicting the reflection parameter from training data as done in Ultra-NeRF, the proposed method explicitly estimates the geometrical normal via physical priors, yielding more accurate geometry-aware 3D reconstructions and producing more precise anisotropic synthesis from different viewpoints. Numerical experiments are conducted and presented in Fig. 5 to validate the effectiveness of the proposed method in detecting the geometric interface between tissues.

The scattering intensity is expressed as the product of the residual wave amplitude and the convolution of the point spread function (PSF) $H(l)$ with the distribution of random scatterers similar to [27],

$$B(\mathbf{r}, t) = I(\mathbf{r}, t) * H(\mathbf{r}, t) \otimes T(x, y, z) \quad (13)$$

where $T(x, y, z)$ represents the scatterers, sampled from a Bernoulli distribution based on the scattering density ρ , with their scattering intensity controlled by the scattering coefficient ϕ .

$$T = \text{Bernoulli}(\rho(\mathbf{r}, t)) * \phi(\mathbf{r}, t) \quad (14)$$

Since ultrasound scattering is caused by tiny particles within tissues, we propose to use the fine-scale hash encoding features $h_{\Omega_2}(\mathbf{x})$, along

with the geometry feature $\mathbf{g}$, to predict the parameters α, φ and ρ as defined in (13).

$$(\alpha, \varphi, \rho) = f_{\Theta_2}(h_{\Omega_2}(\mathbf{x}), \mathbf{g}) \quad (15)$$

Finally, the geometry-aware ultrasound image is generated by the combination of reflection and scattering, as described in (3).

3.3. Loss function

The loss function consists of three components, which can be expressed as:

$$\mathcal{L} = \mathcal{L}_2 + \lambda_1 \mathcal{L}_{MS-SSIM} + \lambda_2 \mathcal{L}_{Scatter} \quad (16)$$

where $\mathcal{L}_2 = \|E - I\|^2$ represents the mean square error between the reconstructed image E and the groundtruth ultrasound image I , $\mathcal{L}_{MS-SSIM} = MS-SSIM(E, I)$ is the multi-scale structural similarity index measure [35] improving human perception of image quality, and $\mathcal{L}_{Scatter} = \sum_{i \in I} \phi(i)^2$ is a regularization term used to constrain the scattering coefficient. λ_1 and λ_2 are hyperparameters. The objective function combines $\mathcal{L}_2$ to ensure pixel-level fidelity and $\mathcal{L}_{MS-SSIM}$ with a weight of $\lambda_1 = 9$ to emphasize perceptual structural details, which is crucial for preserving structural consistency to assist medical diagnosis. For the regularization term $\mathcal{L}_{Scatter}$, we assign a small weight λ_2 (0.0005 – 0.001). This serves as a supplementary safeguard to prevent the scattering component from overfitting to reflection boundaries, thereby enhancing the inherent decoupling capability of our architecture.

4. Experiments

In this section, we first introduce the datasets, metric and implementation details used in the proposed method. Then qualitative and quantitative experiments are conducted to demonstrate the advantages of our method in reconstruction fidelity, reconstruction speed, interpretability and view-dependency. Finally, two ablation study experiments are conducted to validate the effectiveness of the proposed methods.

4.1. Experimental settings

Datasets. We conduct ultrasound reconstruction experiments on two datasets. The first is a synthetic liver dataset provided by Ultra-NeRF [11], which includes ultrasound sweeps collected from six different angles, each sweep consisting of 200 ultrasound images. We use four of these sweeps for training and validate our reconstruction results on the remaining two sweeps. The second dataset provided by [36] captures a sequence of tracked ultrasound images of a pork loin, with an embedded hydrogel sphere that serves as a simulated lesion. We randomly select 3/4 of the frames from this sweep for reconstruction and validate the reconstruction performance on the remaining frames.

Metrics. In this paper, several evaluation metrics are utilized to demonstrate the effectiveness of our method. Specially, we adopt the structural similarity index (SSIM), the root mean square error (RMSE) and the learned perceptual image patch similarity (LPIPS) [37] to compare the errors between reconstructed image and ground truth. SSIM measures the similarity of image structure, RMSE defines the difference of image pixel, and LPIPS measures the average feature distances between two images from a global view, which is very close to human visual perception.

Implementation details. We train our model using the Adam optimization algorithm on an NVIDIA GTX 2080Ti GPU. For both the liver and pork datasets, we extract sampling points that correspond directly to the pixel points of the training and testing images. The iteration number for train is 10,000 except for reconstruction speed experiments. Our network employs 16-level hash encodings, utilizing ReLU-based multi-layer perceptron with six hidden layers of size 64 for f_{Θ_1} and four hidden layers of size 64 for f_{Θ_2} .

Table 1
Quantitative comparisons between our method and other methods.

Dataset	Metrics	Ours			Ultra-NeRF			Nerf			VNN		
		mean	min	max	mean	min	max	mean	min	max	mean	min	max
Liver	SSIM↑	0.49	0.42	0.67	0.44	0.38	0.61	0.54	0.47	0.70	0.39	0.34	0.54
	RMSE↓	0.053	0.045	0.062	0.070	0.059	0.081	0.046	0.040	0.055	0.068	0.049	0.08
	LPIPS↓	0.20	0.17	0.24	0.22	0.18	0.25	0.45	0.32	0.51	0.23	0.18	0.28
Pork	SSIM↑	0.61	0.56	0.69	0.53	0.51	0.57	0.67	0.64	0.70	0.48	0.46	0.55
	RMSE↓	0.060	0.047	0.070	0.078	0.064	0.085	0.056	0.047	0.061	0.10	0.073	0.12
	LPIPS↓	0.18	0.13	0.24	0.24	0.21	0.28	0.33	0.29	0.37	0.26	0.18	0.37

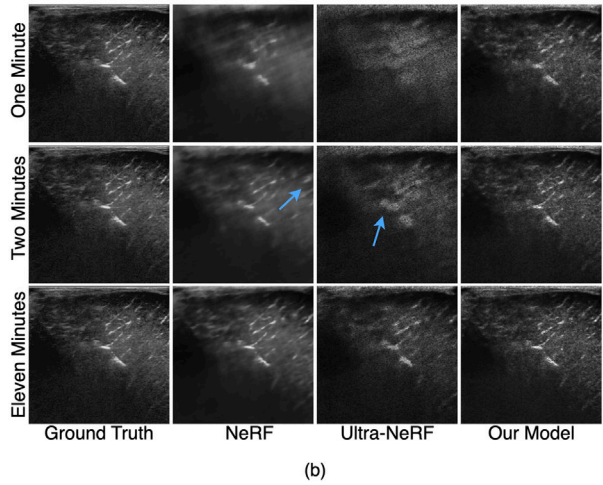
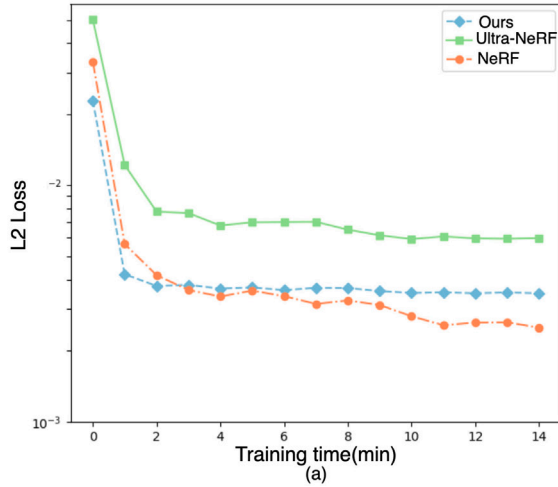


Fig. 4. Comparison of our method, Ultra-NeRF and NeRF in terms of reconstruction speed. (a) the change of reconstruction loss of test view synthesis images during training process. (b) Examples of images synthesis from test views after 1 minute, 2 minutes, and 11 minutes training.

4.2. Experimental results

In this section, we qualitatively and quantitatively evaluate the performance of the proposed method and the following methods: the original NeRF [6], the Ultra-NeRF [11] and the popular explicit reconstruction algorithm voxel nearest neighbor interpolation (VNN) [2].

4.2.1. Reconstruction quality

Fig. 3 qualitatively illustrates the reconstruction results of test views among our method, Ultra-NeRF, NeRF and VNN. Fig. 3(a) and (c) are the generated ultrasound images from new viewpoints on the pork and liver datasets, respectively. The enlargement of partial region marked in red are shown in Fig. 3(b) and (d). We can see that our method can capture more tissue details compared to other methods, such as the surface of hydrogel ball in Fig. 3(b). This improvement can be attributed to the multi-scale hash encoding strategy, which facilitates the extraction of features across multiple levels. Furthermore, by explicitly modeling the geometric relationship between acoustic impedance and tissue interfaces, our method produces significantly clearer reflection surfaces than Ultra-NeRF. Meanwhile, we also observe that the original NeRF exhibits an oversmoothing effect, leading to the loss of important scattering-related details, while VNN suffers from discontinuities due to its reliance on discrete voxel grids.

Table 1 provides a detailed quality evaluation for new view image generation, where SSIM [38], RMSE and LPIPS [37] metrics are used to measure the accuracy between the rendered image and the corresponding ground truth. As shown in Table 1, our method significantly outperforms Ultra-NeRF and VNN in terms of SSIM and RMSE metrics. This demonstrates that our method achieves good accuracy in both image structure and pixel-level errors. Although NeRF obtains the lowest error in SSIM and RMSE metrics, it generates overly smooth rendered images (seen in Fig. 3). The main reason is that ultrasound images contain

complex textures, reflections, and scattering, which are unsuitable for pixel-level metrics to assess accurately. In addition, original NeRF lacks a foundation in ultrasound physics, leading to the loss of scattering information. For the LPIPS metric, our method achieves the best performance and is the closest to human visual perception. As LPIPS metric is learned by neural networks and is well-suited for capturing subtle visual details [39], we believe it provides a more accurate measure of the clinical relevance and visual quality of reconstructed ultrasound images.

4.2.2. Convergence and stability

Although recent implicit representation methods have made significant progress in ultrasound reconstruction, the convergence speed and stability of these methods present a major obstacle to their practical application in medical settings. Our method significantly improves the model's convergence speed and stability based on the two aspects. On the one hand, the use of separable multi-scale hash encoding guides the network to be trained by a coarse-to-fine strategy, enabling faster fitting of tissue parameters and accelerating the convergence of our methods. On the other hand, the proposed physics-based volume rendering model adopting the relation between acoustic impedance and tissue surface demonstrates good stability compared to previous implicit ultrasound reconstruction methods.

Fig. 4(a) illustrates the above analysis of our model, Ultra-NeRF, and NeRF under the mean square error between the corresponding generated images and the ground truth during the training process. It is obvious that the mean square error of all these three models decreases rapidly within the first two minutes and subsequently exhibits fluctuations around a stable level. However, the difference lies in that our method reaches near convergence by the second minute, while Ultra-NeRF continues to adjust slowly and only fully converges around ten minutes, and NeRF takes at least 11 minutes. Fig. 4(b) further presents

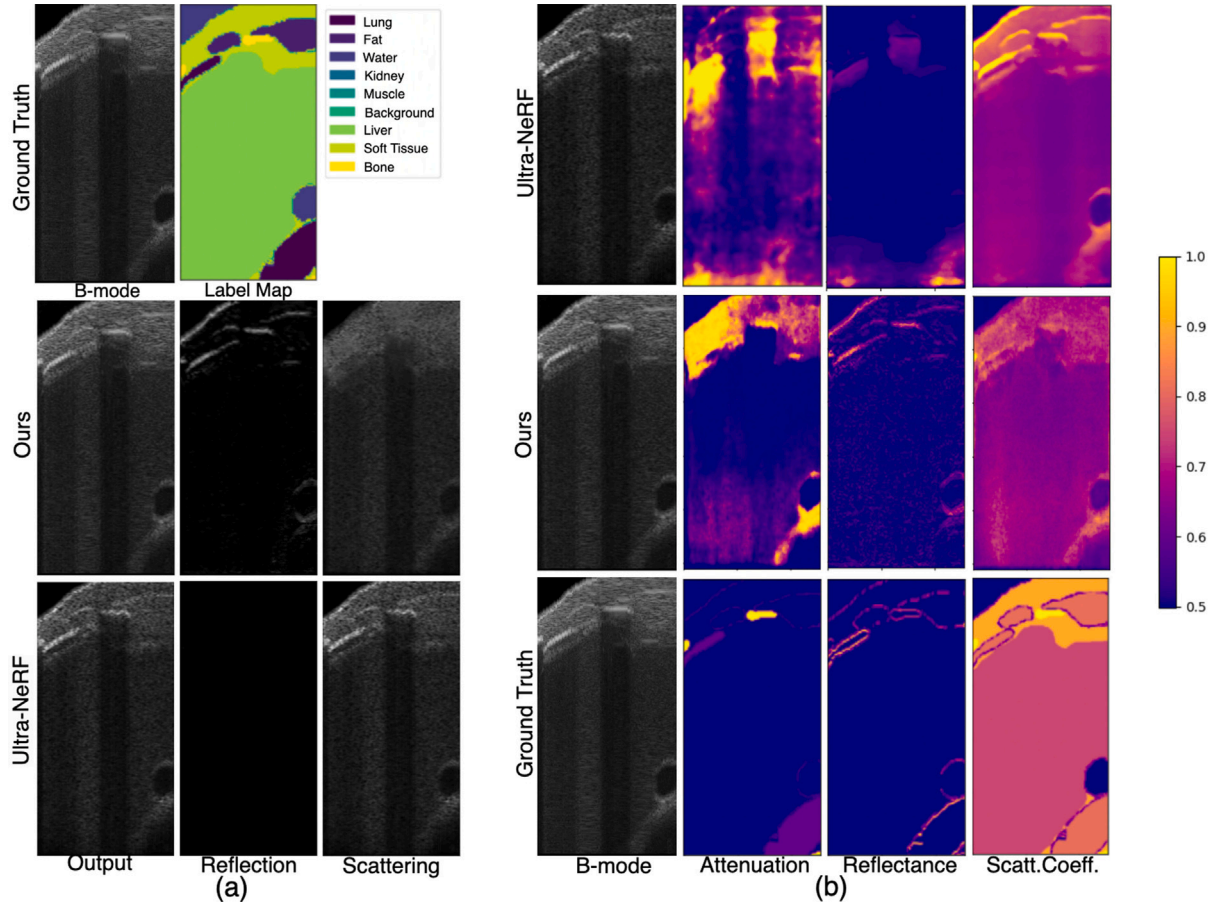


Fig. 5. The effectiveness of the proposed geometry-related constraint in distinguishing between reflections from large tissue interfaces and scattering from microstructures.

the reconstruction results during training. It can be seen that, our model have already achieved accurate reconstruction during the first two minutes. As a contrast, Ultra-NeRF and NeRF only learn the rough contours of tissues at the beginning and fail to capture finer details. For example, Ultra-NeRF does not capture the tissue detail after two minutes and the generated results by NeRF is very blur, as can be seen in the area pointed by the blue arrows.

4.2.3. Geometry-related reflection

In this section, we will demonstrate the effectiveness of the proposed geometry-related constraint in distinguishing between reflections from large tissue interfaces and scattering from microstructures. Fig. 5(a) first illustrates the decomposition results of ultrasound reflections and scattering reconstructed by our method and Ultra-NeRF. As Ultra-NeRF ignores the strong physical correlation between reflections and the geometry of human tissues, the reconstruction results are almost entirely dominated by scattering components in Fig. 5(a). In contrast, our model not only captures the geometric information of human tissues more effectively but also decouples multi-scale features, which helps distinguish broader reflections from fine-grained scattering. Furthermore, compared with the tissue labels of ground truth in Fig. 5(a), it is obvious that the proposed method can precisely capture the reflections occurring at the tissue interfaces and the low-intensity scattering points within the tissues.

In order to further demonstrate the performance of the proposed geometry-aware view-dependent 3D ultrasound reconstruction, Fig. 5(b) shows the distribution of tissue parameters fitted by different methods, such as the attenuation coefficient α , the reflectance β , and the scattering coefficient φ . Compared to Ultra-NeRF, the proposed model has better

ability in fitting the above tissue parameters. For example, our method captures the strong physical correlation between ultrasound reflections and the geometry interface of human tissue by fitting the acoustic impedance of tissue in the second column of Fig. 5(b). In addition, Ultra-NeRF incorrectly learns the bright reflections at the boundaries when fitting the scattering coefficient, as shown in the third column of Fig. 5(b).

As can be seen in Fig. 5, our method can capture the relation between reflections and tissue geometry more accurately, while the extraction of multi-scale information enables us to capture more complex details. Therefore, our method can achieve higher quality and precision in novel views generation.

4.2.4. View dependency for ultrasound physics

As the angle between the ultrasound beam and the tissue interface is taken into account in the reflectance calculation in (11), our method can achieve anisotropic 3D ultrasound reconstruction that current ultrasound reconstruction techniques based on deep learning had always failed. We believe that the modeled view-dependent behavior stems from the synergy between our explicit physics-based modeling and the implicitly learned acoustic impedance field. Specifically, the mechanism of view-dependence is explicitly governed by (11), which enforces Lambert's cosine law using the dot product between the viewing direction $\mathbf{d}$ and the surface normal $\mathbf{n}$. The geometry driving this model (the surface normal $\mathbf{n}$) is derived from the gradient of the implicitly learned acoustic impedance field ($\nabla_x z$). Fig. 6 illustrates the reconstruction results from different views on the same tissue cross-section. The ultrasound incident angles from left to right are 30° , 5° , 0° , -5° , and -30° , respectively. By carefully observing these reconstructed

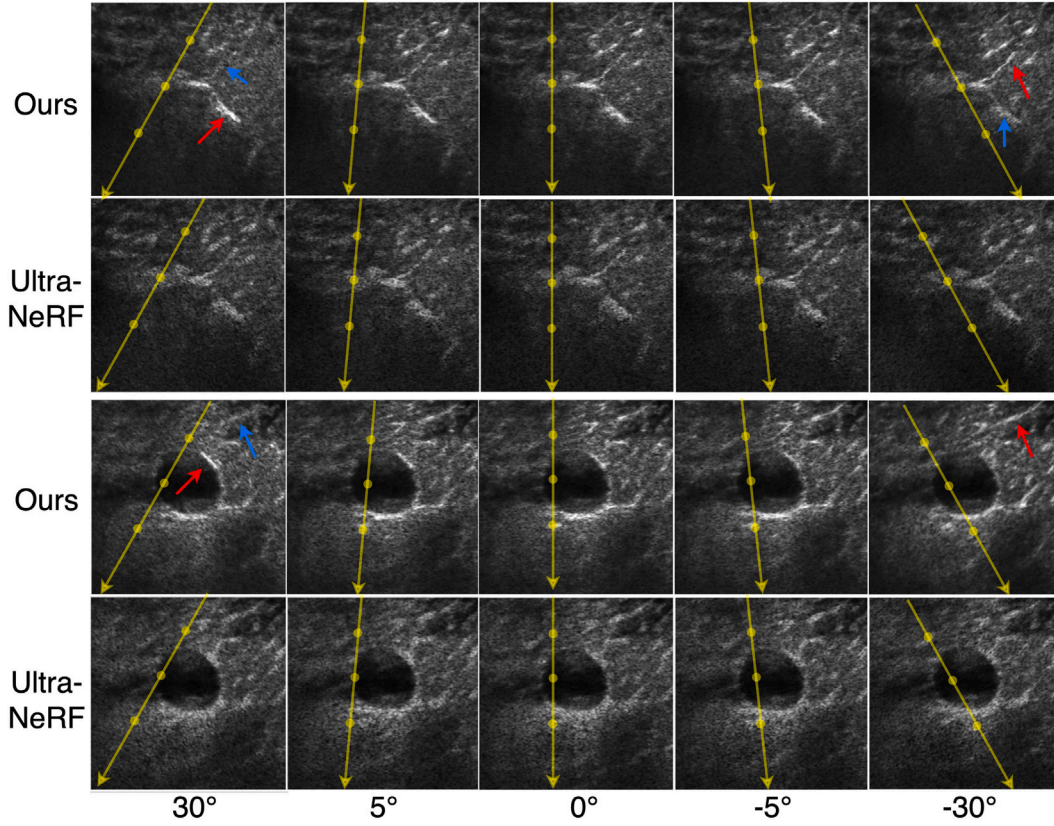


Fig. 6. Comparisons on the view dependency for ultrasound physics on the same cross-section under different ultrasound incident angles.

Table 2

Quantitative comparison between Unified and separable encoding strategies. Our separable design achieves superior performance across all metrics.

Strategy	SSIM↑	RMSE↓	LPIPS↓
Unified	0.47	0.060	0.21
Separable (Ours)	0.49	0.053	0.20

ultrasound images under different incident angles, we find that the proposed method can effectively reflect the physical mechanism of ultrasound reconstruction, as the angle between the ultrasound beam and the normal to the tissue surface increases, the reflection intensity diminishes and may even vanish. For example, the surface of the hydrogel beads or the fascia within the muscle pointed by the red arrows show great brightness as the angle between the ultrasound beam and the normal of tissue surface approximates 0° . Conversely, the area pointed by the blue arrow is dimmed because it is parallel to the direction of the ultrasound beam. As a contrast, Ultra-NeRF can not deal with the intrinsic property of ultrasound images.

4.3. Ablation study

In this section, several experiments are conducted to further demonstrate the enhanced efficacy in 3D ultrasound reconstruction.

4.3.1. Effectiveness of separable encoding strategy

As introduced in Section 3.1, we employ a separable multi-scale hash encoding strategy to explicitly decouple the feature extraction for reflection and scattering. To validate the necessity of this design, we compare our method against a “Unified” strategy where all hash levels are fed into both networks simultaneously, defined as:

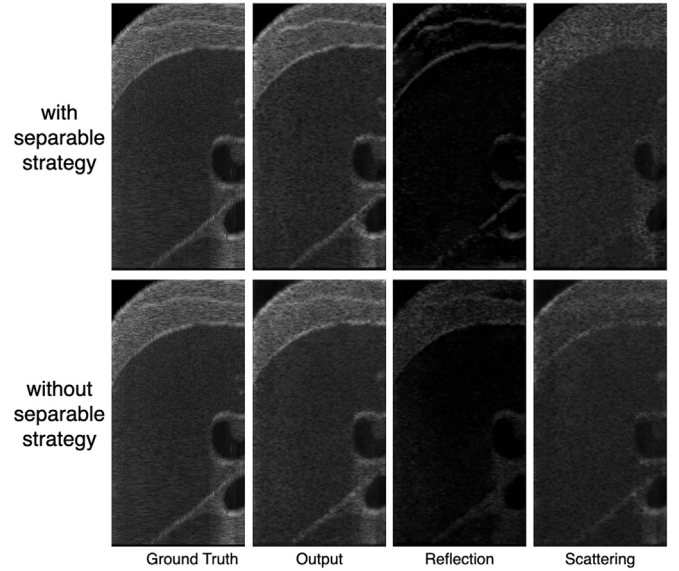


Fig. 7. Visual comparison between Unified and separable encoding strategies. The Unified strategy (top) suffers from noise leakage in the reflection component, whereas our separable strategy (bottom) successfully disentangles smooth tissue boundaries from fine-grained scattering textures.

$$(z, \mathbf{g}) = f_{\Theta_1}(h_{\Omega}(\mathbf{x})), \quad (\alpha, \varphi, \rho) = f_{\Theta_2}(h_{\Omega}(\mathbf{x}), \mathbf{g}). \quad (17)$$

Table 2 reports the quantitative performance of both strategies. It is evident that the proposed separable strategy yields consistent improvements across all metrics (SSIM, RMSE, and LPIPS). This quantitative advantage is deeply rooted in the model’s ability to handle physical constraints. Visual comparisons in Fig. 7 further corroborate

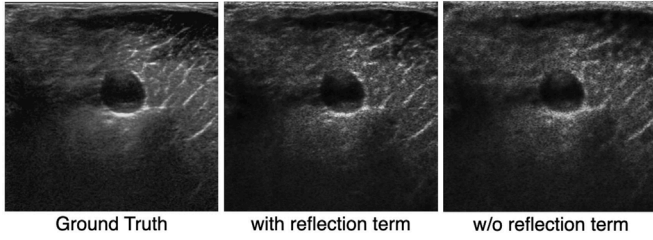


Fig. 8. Ablation study on ultrasound reflection term.

Table 3

Quantitative comparisons on whether to use ultrasound reflection.

	With reflection term	W/o reflection term
SSIM↑	0.61	0.56
RMSE↓	0.060	0.069
LPIPS↓	0.18	0.20

these findings. The unified strategy suffers from *feature entanglement*, where the network fails to discriminate between low-frequency geometric structures and high-frequency speckle noise. Consequently, the reflection component in the unified model appears noisy and contaminated by scattering textures. In contrast, our separable design enforces a strong *physical prior* by restricting the acoustic impedance learning to coarse hash levels. This acts as a beneficial *Inductive Bias*, effectively preventing the reflection network from overfitting to high-frequency noise. Consequently, our method produces cleaner, more physically plausible tissue interfaces, demonstrating a superior capability to decouple geometry from texture. Furthermore, regarding the final rendering results, our method exhibits superior representational capacity. By correctly assigning high-frequency details to the scattering component and low-frequency geometry to the reflection component, the final synthesized image retains sharp structural edges without losing the delicate speckle patterns characteristic of ultrasound images.

4.3.2. The influence of ultrasound reflection

To validate the role of the proposed ultrasound reflection in (3), we conduct a contrast experiment on the pork dataset by reconstructing images using only the ultrasound scattering term, written as

$$E(r, t) = B(r, t). \quad (18)$$

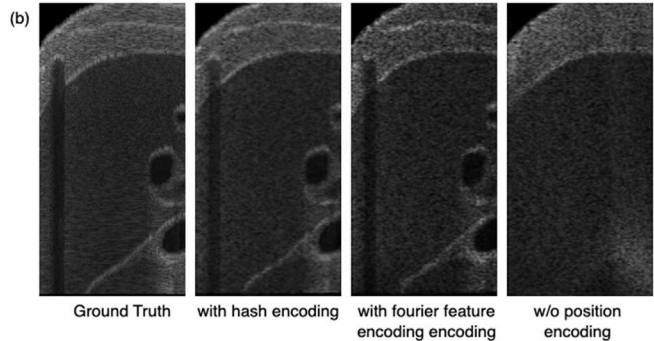
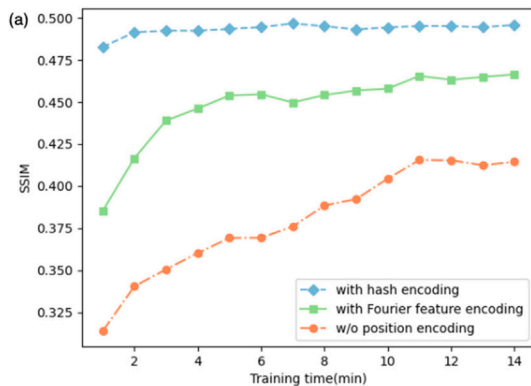


Fig. 9. Ablation study on different positional encoding methods. We compare the reconstruction performance using (i) No positional Encoding, (ii) Fourier-based positional Encoding, and (iii) our multi-resolution hash Encoding. Subfigure (a) illustrates the convergence of SSIM over training time, demonstrating the superior efficiency of our method. Subfigure (b) visualizes the final reconstruction results, where our method captures finer high-frequency details compared to the baselines.

As illustrated in Fig. 8, compared to the ground truth, the reconstruction results without ultrasound reflection loss many tissue details, especially those on tissue surfaces. Table 3 further shows the average error on the SSIM, RMSE, and LPIPS metrics between reconstructed images and the ground truth for the two methods on the test pork dataset. We can find that these three measures have been significantly improved after ultrasound reflection term is considered. Hence, the proposed ultrasound reflection is very effective for 3D ultrasound reconstruction.

4.3.3. Impact of positional encoding

To further justify the motivation for adopting multi-resolution hash encoding, we conducted a specific ablation study comparing three different positional encoding methods: (i) no positional encoding (using raw coordinates), (ii) Fourier-based positional encoding (as used in standard NeRF [6]), and (iii) our proposed multi-resolution hash encoding. As illustrated in Fig. 9, the choice of positional encoding significantly impacts both reconstruction quality and convergence speed. Specifically, regarding *reconstruction quality*, the method without positional encoding fails to capture high-frequency details due to the “spectral bias” of MLP neural networks, resulting in overly smooth and blurry reconstructions as shown in Fig. 9(b). While Fourier-based positional encoding improves the ability to represent details, its reconstruction results remain relatively coarse, struggling to resolve the fine-grained speckle patterns inherent in ultrasound images. In contrast, utilizing the multi-resolution characteristics of hash encoding, our method reconstructs the finest scattering details and the sharpest tissue boundaries. Regarding *convergence speed*, as shown in Fig. 9(a), our method achieves significantly faster convergence and superior convergence results on the SSIM metric compared to the Fourier baseline, demonstrating the efficiency and effectiveness of the hash table query mechanism; this advantage is even more pronounced compared to the method without positional encoding.

5. Conclusion

In this paper, we introduce a geometry-aware view-dependent ultrasound representation method that learns a deep neural implicit representation of a three-dimensional object from a set of two-dimensional ultrasound frames. The proposed model is grounded in ultrasound physics and uses separable multi-scale hash encoding features to distinguish and represent anisotropic broad ultrasound reflections and fine-grained scattering, which is very effective for 3D ultrasound reconstruction. Moreover, the proposed method demonstrates notable improvements over existing techniques both qualitatively and quantitatively.

Despite our method achieves significant advancements, there are still some limitations. One limitation lies in the assumption of a constant

speed of sound and straight-line propagation, inherited from the convolutional ray-tracing model [27]. Consequently, our current framework does not account for the geometric distortions caused by ultrasound refraction at tissue interfaces with varying acoustic velocities. In addition, identifying ultrasound artifacts [25] in ultrasound imaging during the reconstruction process remains a challenge. These issues are left for our future research.

CRedit authorship contribution statement

Bin Liu: Writing – review & editing, Supervision, Methodology, Funding acquisition. **Jiahao Kang:** Visualization, Validation, Project administration, Investigation, Data curation. **Zizheng Li:** Visualization, Data curation. **Tao Gu:** Investigation, Data curation. **Zhi-Fen He:** Resources, Funding acquisition, Formal analysis. **Congxuan Zhang:** Resources, Formal analysis. **Bo Li:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] D.G. Gobbi, T.M. Peters, Interactive Intra-Operative 3D Ultrasound Reconstruction and Visualization, MICCAI, Tokyo, Japan, 2002, pp. 156–163. https://doi.org/10.1007/3-540-45787-9_20
- [2] S. Sherebrin, A. Fenster, R.N. Rankin, D. Spence, Freehand three-dimensional ultrasound: implementation and applications, SSPIE 2708, Medical Imaging 1996: Physics of Medical Imaging (1996), <https://doi.org/10.1117/12.237790>
- [3] T. Wen, F. Yang, J. Gu, L. Wang, A novel bayesian-based nonlocal reconstruction method for freehand 3D ultrasound imaging, Neurocomputing 168 (2015) 104–118.
- [4] Q. Huang, Z. Zeng, A review on real-time 3D ultrasound imaging technology, BioMed Res. Int. 1 (2017) 6027029, <https://doi.org/10.1155/2017/6027029>
- [5] T. Wen, F. Yang, J. Gu, S. Chen, L. Wang, Y. Xie, An adaptive kernel regression method for 3d ultrasound reconstruction using speckle prior and parallel GPU implementation, Neurocomputing 275 (2018) 208–223.
- [6] B. Mildenhall, P.P. Srinivasan, M. Tancik, J.T. Barron, R. Ramamoorthi, R. Ng, NeRF: representing scenes as neural radiance fields for view synthesis, Commun. ACM 65 (1) (2021) 99–106.
- [7] P.-H. Yeung, L. Hesse, M. Aliasi, M. Haak, W. Xie, A.I. Namburete, et al., ImplicitVol: Sensorless 3d ultrasound reconstruction with deep implicit representation, arXiv preprint arXiv:2109.12108, 2021, <https://doi.org/10.48550/arXiv.2109.12108>
- [8] A.N. Gu, P. Abolmaesumi, C. Luong, K.M. Yi, Representing 3D ultrasound with neural fields, in: Medical Imaging with Deep Learning (MIDL), 2022, <https://openreview.net/forum?id=1EDRk-CyQou>
- [9] H. Li, H. Chen, W. Jing, Y. Li, R. Zheng, 3D ultrasound spine imaging with application of neural radiance field method, in: IEEE International Ultrasonics Symposium (IUS), 2021, pp. 1–4. <https://doi.org/10.1109/ius52206.2021.9593917>
- [10] S. Song, Y. Huang, J. Li, M. Chen, R. Zheng, Development of implicit representation method for freehand 3D ultrasound image reconstruction of carotid vessel, in: IEEE International Ultrasonics Symposium (IUS), 2022, pp. 1–4. <https://doi.org/10.1109/IUS54386.2022.9958448>
- [11] M. Wysocki, M.F. Azampour, C. Eilers, B. Busam, M. Salehi, N. Navab, Ultra-nerf: neural radiance fields for ultrasound imaging, in: Medical Imaging with Deep Learning, PMLR, 2024, pp. 382–401.
- [12] H. Fan, H. Su, L. Guibas, A Point Set Generation Network for 3d Object Reconstruction from a Single Image, CVPR, In, 2017, <https://doi.org/10.1109/cvpr.2017.264>
- [13] P. Mandikal, K. Navaneet, M. Agarwal, R.V. Babu, 3D-LMNet: Latent embedding matching for accurate and diverse 3D point cloud reconstruction from a single image, arXiv preprint arXiv:1807.07796, 2018, <https://doi.org/10.48550/arXiv.1807.07796>
- [14] N. Wang, Y. Zhang, Z. Li, Y. Fu, W. Liu, Y.-G. Jiang, Pixel2Mesh: generating 3D mesh models from single RGB images, in: European Conference on Computer Vision (ECCV), 2018, pp. 55–71, https://doi.org/10.1007/978-3-030-01252-6_4
- [15] C. Wen, Y. Zhang, Z. Li, Y. Fu, Pixel2Mesh++: 3D mesh generation and refinement from multi-view images, ICCV (2019) 2019, <https://doi.org/10.1109/iccv.2019.00113>
- [16] A. Kar, C. Häne, J. Malik, Learning a multi-view stereo machine, Neural Inf. Process. Syst. 30 (Jan 2017).
- [17] C.B. Choy, D. Xu, J. Gwak, K. Chen, S. Savarese, 3D-R2N2: a unified approach for single and multi-view 3D object reconstruction, in: European Conference on Computer Vision (ECCV), 2016, pp. 628–644, https://doi.org/10.1007/978-3-319-46484-8_38
- [18] H. Xie, H. Yao, X. Sun, S. Zhou, S. Zhang, Pix2Vox: context-aware 3D reconstruction from single and multi-view images, ICCV pp. 2690–2698 (2019), <https://doi.org/10.1109/iccv.2019.00278>
- [19] J.J. Park, P. Florence, J. Straub, R. Newcombe, S. Lovegrove, Deepsdf: learning continuous signed distance functions for shape representation, CVPR pp. 165–174 (2019), <https://doi.org/10.1109/cvpr.2019.00025>
- [20] S. Lombardi, T. Simon, J. Saragih, G. Schwartz, A. Lehrmann, Y. Sheikh, Neural volumes: learning dynamic renderable volumes from images, ACM Trans. Graph. (2019) 1–14, <https://doi.org/10.1145/3306346.3323020>
- [21] M. Niemeyer, L. Mescheder, M. Oechsle, A. Geiger, Differentiable volumetric rendering: learning implicit 3D representations without 3D supervision, in: CVPR, 2020, <https://doi.org/10.1109/cvpr42600.2020.00356>
- [22] S. Fridovich-Keil, A. Yu, M. Tancik, Q. Chen, B. Recht, A. Kanazawa, Plenoxels: radiance fields without neural networks, CVPR pp. 5501–5510(2022), <https://doi.org/10.1109/cvpr52688.2022.00542>
- [23] C. Sun, M. Sun, H.-T. Chen, Direct voxel grid optimization: super-fast convergence for radiance fields reconstruction, CVPR pp. 5459–5469 (2022), <https://doi.org/10.1109/cvpr52688.2022.00538>
- [24] T. Müller, A. Evans, C. Schied, A. Keller, Instant neural graphics primitives with a multiresolution hash encoding, ACM Trans. Graph. (2022) 1–15, <https://doi.org/10.1145/3528223.3530127>
- [25] J. Zagzebski, Essentials of Ultrasound Physics, Mosby, 1996, <https://books.google.com.hk/books?id=XahsQgAACAAJ>
- [26] F. Kremkau, Diagnostic Ultrasound: Principles and Instruments, W.B. Saunders, 2002, <https://books.google.com/books?id=TiBAAAAAAAJ>
- [27] M. Salehi, S.-A. Ahmadi, R. Prevost, N. Navab, W. Wein, Patient-specific 3d ultrasound simulation based on convolutional ray-tracing and appearance optimization, in: MICCAI 2015, Springer, 2015, pp. 510–518, https://doi.org/10.1007/978-3-319-24571-3_61
- [28] J.W. Trobaugh, D.J. Trobaugh, W.D. Richard, Three-dimensional imaging with stereotactic ultrasonography, Comput. Med. Imaging Graph. 18 (5) (1994) 315–323, [https://doi.org/10.1016/0895-6111\(94\)90002-7](https://doi.org/10.1016/0895-6111(94)90002-7)
- [29] Q. Huang, Y. Huang, W. Hu, X. Li, Bezier interpolation for 3-D freehand ultrasound, IEEE Trans. Human-Machine Syst. (2015) 385–392, <https://doi.org/10.1109/thms.2014.2374551>
- [30] J.M. Sanches, J.S. Marques, A multiscale algorithm for three-dimensional free-hand ultrasound, Ultrasound Med. Biol. 28 (8) (2002) 1029–1040, [https://doi.org/10.1016/s0301-5629\(02\)00548-3](https://doi.org/10.1016/s0301-5629(02)00548-3)
- [31] A. Corona-Figueroa, J. Frawley, S. Bond-Taylor, S. Bethapudi, H.P. Shum, C.G. Willcocks, MedNeRF: medical neural radiance fields for reconstructing 3D-aware CT-projections from a single X-ray, in: IEEE Engineering in Medicine & Biology Society (EMBC), 2022, pp. 3843–3848, <https://doi.org/10.1109/EMBC48229.2022.9871757>
- [32] H. Li, J. Yang, M. Qu, Y. Feng, A semi-supervised multi-task assisted method for ultrasound medical image segmentation, Neurocomputing (2025) 130217.
- [33] N. Zaidkilani, M.A. Garcia, D. Puig, CoAtUNet: a symmetric encoder-decoder with hybrid transformers for semantic segmentation of breast ultrasound images, Neurocomputing 629 (2025) 129660.
- [34] W. Wein, S. Brunke, A. Khamene, M.R. Callstrom, N. Navab, Automatic ct-ultrasound registration for diagnostic imaging and image-guided intervention, Med. Image Anal. 12 (5) (2008) 577–585, <https://doi.org/10.1016/j.media.2008.06.006>
- [35] Z. Wang, E.P. Simoncelli, A.C. Bovik, Multiscale structural similarity for image quality assessment, in: The Thirty-Seventh Asilomar Conference on Signals, Systems Computers, 2003, Vol. 2, 2003, pp. 1398–1402, <https://doi.org/10.1109/ACSSC.2003.1292216>
- [36] A.M. Wijata, M. Biesok, J. Juszczak, J. Nalepa, A phantom lesion dataset for precise estimation of lesion volume from ultrasound images, 2023. <https://doi.org/10.7910/DVN/OSYZUH>
- [37] R. Zhang, P. Isola, A.A. Efros, E. Shechtman, O. Wang, The unreasonable effectiveness of deep features as a perceptual metric, Proc. CVPR (2018) 586–595, <https://doi.org/10.1109/CVPR.2018.00068>
- [38] Z. Wang, A.C. Bovik, H.R. Sheikh, E.P. Simoncelli, Image quality assessment: from error visibility to structural similarity, IEEE Trans. Image Process. 13 (4) (2004) 600–612, <https://doi.org/10.1109/TIP.2003.819861>
- [39] Y. Blau, T. Michaeli, The perception-distortion tradeoff, CVPR 6228–6237 (2018), <https://doi.org/10.1109/CVPR.2018.00652>

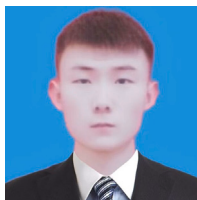
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